

Meet The Great American Star

Student Introduction

The Great American Star is a piece of mathematical art created for America's 250th anniversary. It combines patterns, path-counting, patriotism, and personal storytelling in one star-shaped design.



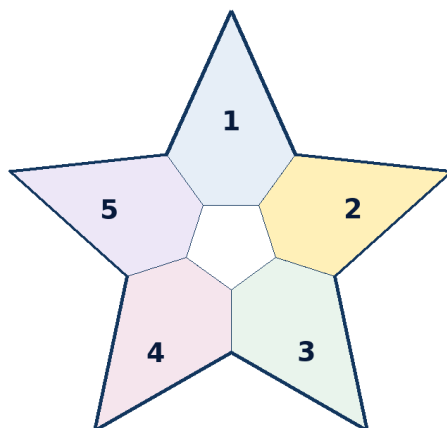
5	repeating sections
50	starting points in the full star
50	ending points in the full star
24	steps in one complete path
359,277,250	possible paths in the full star
6	image spaces to personalize

What students should notice first

The full star is made of **five repeating sections**. Each path begins near the center, travels outward to a tip, and spells **UNITEDSTATESOFAMERICA250**. The star is mathematical, but it can also become personal when you choose your own six images.

One Section, Repeated Five Times

The complete star is built from one section pattern that is repeated five times. If you understand one section, you are already on your way to understanding the whole object.



Same section pattern
repeated 5 times

One section	71,855,450 paths
Five sections	$71,855,450 \times 5 = 359,277,250$ paths

Big idea

There are many possible paths, but they all belong to **one shared American star**. A student can choose one route that is uniquely theirs while still being part of the whole.

A small extra surprise: some corners are shared where sections meet, which is why the full star ends up with **50 distinct starting points** rather than simply 55.

How One Path Works

A complete path begins at one of the starting points near the center and moves outward to one of the tips. Along the way it passes through **24 character positions**. In order, those 24 positions spell:

UNITED STATES OF AMERICA 250

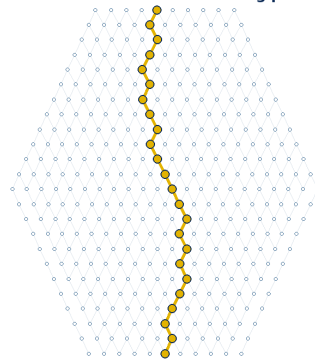
Written without spaces, that is exactly 24 characters: **UNITEDSTATESOFAMERICA250**. Every complete route in the star follows that 24-step phrase.

One complete path = 24 steps / character positions

Center side → one star tip

TIP / END

10 ending positions



Every path spells:

**UNITED
STATES
OF
AMERICA
250**

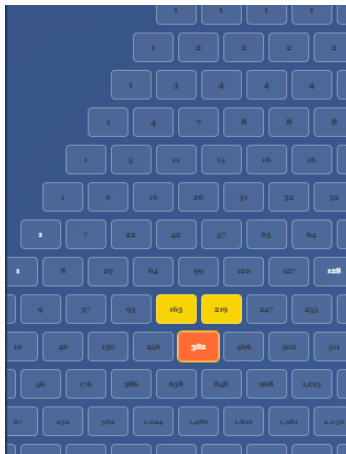
24 rows / positions

The section:

- starts with 11
- widens to 22
- narrows to 10

Gold line = one complete path.

11 starting positions in row 0
CENTER / START



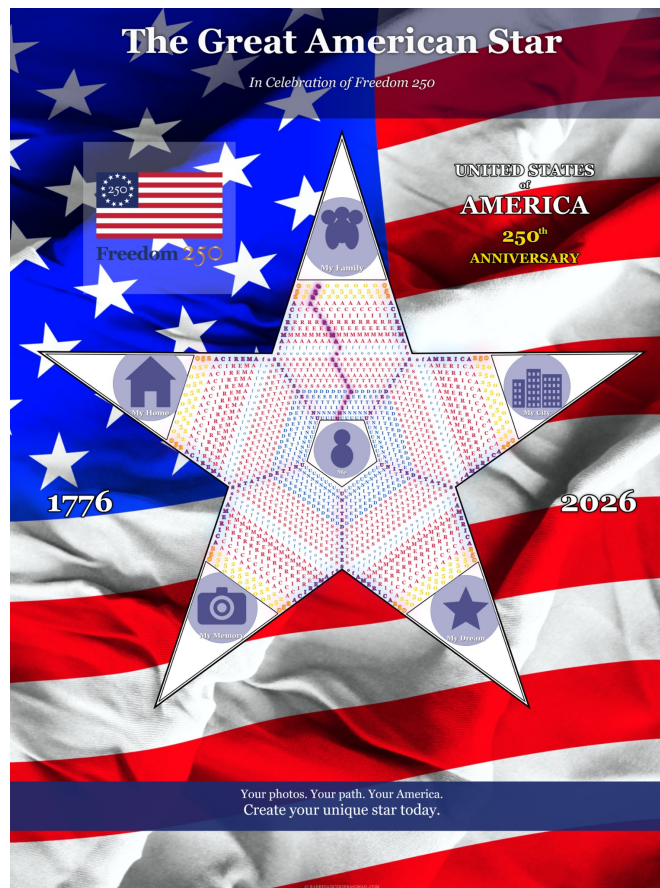
A Modified Pascal's Triangle

Many students already know Pascal's Triangle from school. The Great American Star uses a **modified** version of that idea.

In one section there are **11 starting positions in row 0**. The rows then get wider and wider, then narrower again, ending with **10 ending positions**. The branching logic is Pascal-like, but the shape is adapted to fit a star section.

Make It Personal: Six Image Spaces

The Great American Star is not only a math object. It also lets each student create a personal version of the star by choosing a path and adding six images.



Personalization ideas

You might choose six images about your family, interests, goals, favorite places, or school themes. Schools can also use a set of approved images for the activity.

Privacy note: the images used in the creator stay private. The website owner does not have access to them through the website.

Try these reflection prompts

1. If you made your own star, what six images would you choose?
2. Which part of the star feels most mathematical to you?
3. Which part feels most personal or creative?
4. What do you want to learn next: starting points, path counts, or probability?

Next steps: after understanding the whole object, you can explore deeper lessons such as how different starting points create different numbers of paths, or how probability ideas appear in the Birthday Problem lesson.